Coprime scale carriers and fixed prime separation

Round thirteen research contribution

24 October 2026

1 Coprime scale carriers and fixed prime separation

A low-degree scale and fixed prime separation can be combined without adding a variable. The resulting complete integer equations are

Construction	Variables	Exact degree	Binary operations
Normalized five-root norm	13	1072	192
Ordinary four-root norm	14	378	145
Ordinary three-root norm	15	172	123
Centered three-root norm	15	170	134
Centered two-root norm	16	98	138

All variables range over unrestricted integers. These are explicit cost–degree tradeoffs, not global minima. In particular, the first row improves the incoming 13/1088/195 point; the second substantially lowers both degree and evaluation cost from the inspected 14/652/163 point. The 16-variable row lowers degree but increases cost relative to 16/108/132.

The source is the literal signed CRT parent `degree88_17.json`, with its inherited ordinary native compiler and K215 calibration. We restore each of its five residuals below. This is not a new Lean/Rocq certificate, and it proves neither consistency nor a historical or global optimality claim.

1.1 Complete common prefix

The base thirteen coordinates are

$$j, C, z, h, r, v, w, T, c, n, t, p_2, p_3.$$

The fixed six numerals are the coherent source-34 tuple β, K_0, M_C, M_F, with M_F already shifted, and $A_{\text{cone}} = 2^{1113}$, $B_{\text{cone}} = 2^{666104}$. In particular $\beta = 2^{1064921} - 1$. All named registers below are

polynomial abbreviations, not extra quantified variables:

$$
\begin{aligned}
&J = 6j + 1, && q = \beta J + 1, && W = 12w + 4,\\
&X = qW, && Y = qv, && E = XY,\\
&a = E + Y, && H = 4a + 3, && \Delta = a^2 + H,\\
&F = (K_0 + W)C - z(q-1), &&&&\\
&R = (q(q-F) - C)(q^2-1) + (M_C + qM_F)J, &&&&\\
&k = R + 1 + hE, && \eta = c - Yk, && \zeta = k - \eta,\\
&u = EYk, && A_\tau = u(u+k) + 1, && A_m = 1 + \Delta c^2,\\
&b_0 = X + ac, && I = (2r+1)c^2, && M = 2q^2,\\
&U = MH, && e = v - q^2. &&
\end{aligned}
$$

The determinant and guard radicand are

$$
\begin{aligned}
d &= (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2,\\
A_g &= d\eta\zeta - (t^2 - d - \eta\zeta + 1)^2 - 1 - p_2^2 - p_3(p_3+1).
\end{aligned}
$$

For the ordinary variants, add the coordinate f and set

$$
Q = I^2, \qquad S = Q - \Delta(f^2 - 1), \qquad V = fcT - f^2R - c.
$$

For the thirteen-variable normalized variant instead use

$$
A_f = 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \qquad V = A_f(cT - R) - c.
$$

In both cases put $\delta = Q - 1$, $C_y = QV^2 - 1$. The rational expression $A_y = C_y/\delta$ is proof notation only, never a circuit division. Where the coordinate γ is retained, write $D = (b_0 + H\gamma)^2 - A_m$. Where p_1 is retained, write $G = A_g - p_1^2$.

1.2 Five literal norm specifications

Let

$$
\mathcal{F}_m(x; a_1, \ldots, a_m) = \prod_{\varepsilon \in \{-1,1\}^m} \left(x + \sum_i \varepsilon_i \sqrt{a_i} \right).
$$

Sign invariance proves that this is an integer polynomial. The following expressions are cleared symbolically; their JSON circuits use only binary addition, subtraction and multiplication.

For thirteen variables take $Z = U(n+V) + Mb_0 + He$ and

$$
N_{13} = \delta^{16}\mathcal{F}_5\left(Z; 81U^2A_f, 9U^2A_\tau, M^2A_m, 729U^2A_g, \frac{U^2C_y}{\delta}\right).
$$

The equation is $N_{13} = 0$.

For fourteen variables, add f, retain the same Z, and use

$$
N_{14} = \delta^8\mathcal{F}_4\left(Z; 9U^2A_\tau, 81U^2A_g, M^2A_m, \frac{U^2C_y}{\delta}\right), \qquad P_{14} = (1 + N_{14})(1 + S) - 1.
$$

For fifteen variables, add f, γ, and use

$$N_{15} = \delta^4 \mathcal{F}_3 \left(M(n+V)+e; 9M^2A_\tau, 81M^2A_g, \frac{M^2C_y}{\delta} \right), \qquad P_{15} = (1+N_{15})(1+S)(1+D)-1.$$

The centered variants use $t_0 = 2u+k$, $\ell = k^2-4$. For fifteen variables add f, p_1 and set

$$\begin{aligned} Z_c &= U(n+3t_0+2V)+2Mb_0+He, \\ N_{15c} &= \delta^4 \mathcal{F}_3 \left(Z_c; 36U^2A_\tau, 4M^2A_m, \frac{4U^2C_y}{\delta} \right), \\ P_{15c} &= (1+N_{15c})(1+S)(1+4G)-1. \end{aligned}$$

For sixteen variables add f, γ, p_1, put $x = Mn+e$, and define

$$\begin{aligned} j_0 &= \delta\{x^2 + M(6t_0+4V)x + 9M^2\ell\} - 4M^2(V^2-1), \\ h_0 &= 12\delta t_0 V M^2, \\ N_{16} &= j_0(j_0+2h_0) + 144\delta M^4\{4A_\tau(1-V^2) + \delta\ell V^2\}, \\ P_{16} &= (1+N_{16})(1+S)(1+D)(1+4G)-1. \end{aligned}$$

The exact identity, valid also at $\delta = 0$, is

$$N_{16} = \delta^2 \mathcal{F}_2 \left(M(n+3t_0+2V)+e; 36M^2A_\tau, \frac{4M^2C_y}{\delta} \right)$$

after symbolic clearing. An independent integer-coefficient check compares all 29 nonzero terms in the centered identity.

1.3 Unit products recover the rows before denominator use

For integers, if residuals R_i never equal -2, then

$$(1+N)\prod_i (1+R_i) = 1 \quad \Longleftrightarrow \quad N = R_1 = \cdots = R_s = 0.$$

Indeed every factor of the integer unit 1 is ± 1. Each R_i factor is $+1$, forcing the one unrestricted norm factor to be $+1$ too. There is no missing assumption on the norm modulo four.

Here a is even and $\Delta \equiv 3 \pmod 4$. If f is odd then $S \equiv I^2 \pmod 4$; if f is even then $S \equiv I^2 - 1 \pmod 4$. Thus S is never 2 modulo four. The same is true of D: for even c its residues are $0, 3$, and for odd c they are $0, 1$. Finally $4G \equiv 0 \pmod 4$. All the stated finalizers therefore recover their individual residuals, in particular $S = 0$, before interpreting a rational norm.

Since $a \in 6\mathbb{Z}$, $\Delta = (a+1)(a+3) \geq 3$. In ordinary variants $S = 0$ excludes $Q = 1$, as it would give $\Delta(f^2-1) = 1$. In the normalized variant $Q = \Delta^2 I^2$ excludes $Q = 1$ directly. Hence $\delta \neq 0$ at every zero in question.

1.4 Prime separation before guard recovery

The fixed residues give

$$q \equiv X \equiv 2 \pmod 3, \quad a \equiv H \equiv \Delta \equiv 0 \pmod 3, \quad A_\tau \equiv A_m \equiv 1 \pmod 3.$$

To check A_τ, write $A_\tau = 1 + b(b+1)k^2$ with $b = XY^2 \equiv 0$ or 2 modulo three. Also H is nonzero and odd. Under $S = 0$, the square $Q = I^2$ is divisible by three, hence by nine. The normalized Q is already divisible by nine. Thus A_y is a three-adic unit of residue one.

The weighted roots used by the ordinary fourteen-variable norm have squared orders

$$-2v_3(H), \quad 0, \quad 2, \quad 4 + v_3(A_g)$$

for $d_0/H, y, 3\tau, 9p_1$ respectively. The normalized construction inserts $9f$ of squared order four and uses $27p_1$ of squared order at least six. A zero guard root is harmless.

Here is the precise separation argument. In any subset of at least two nonzero roots, the smallest squared order is unique and even. Divide out its power of three and multiply the putative zero sum over all signs of the remaining roots. The product is a polynomial in the rational squares and is a nonzero unit modulo three, a contradiction. Subtracting a Galois conjugate of a rational signed root sum would give exactly such a forbidden signed subsum. Every root is therefore rational. This argument allows negative radicands; their rationality subsequently rules that case out. No positivity guard is assumed beforehand.

Roots of integer squares are integral. For the auxiliary root, $Q = 0$ gives $y = \pm 1$. For $Q > 1$, rational y in $QV^2 - (Q-1)y^2 = 1$ is integral by adjacent-Pell descent. In lowest terms $y = A/m$ gives $m^2 \mid Q - 1$. The transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves the norm and the denominator, and after taking positive signs satisfies $0 < V' < V$ when $V > 1$. The endpoint $|V| = 1$ forces $m = 1$. Thus the case $c = 0$, and consequently $Q = 0$, has not been silently excluded.

1.5 Coprime carriers and exact source restoration

Because $Y = qv$, we have $H \equiv 3 \pmod{q}$. Since $q \equiv 2 \pmod{3}$ and H is odd,

$$\gcd(H, 2q^2) = 1.$$

After root recovery the ordinary fourteen-variable factor is

$$U(n+V) + Mb_0 + He = Md_0 + U(3\tau + y + 9p_1).$$

Reduction modulo M and H gives respectively

$$v = q^2(2v_{\text{old}} + 1), \qquad \gamma = \frac{d_0 - b_0}{H} \in \mathbb{Z}.$$

The normalized case adds $9f$ and replaces $9p_1$ by $27p_1$; the same argument applies. The centered main carrier instead has $2M(d_0 - b_0)$ modulo H, still invertible because H is odd. The variants retaining γ need only the first congruence. In all cases $Y = (2v_{\text{old}} + 1)q^3$ exactly, including its odd quotient. The removed v^2 guard multiplier is unnecessary: $v \neq 0$ follows now from the recovered congruence, not from an assumed inequality.

The integral guard root restores $G = 0$, which implies $d > 0$ and $\eta\zeta > 0$ by the standard joint-positivity identity. For completeness, if both factors were negative their magnitudes $b, e > 0$ would require $(t^2 + b + e + 1)^2 < be$, which is impossible. Thus $c = 0$ is impossible too, since it would give $\eta\zeta = -Y(Y+1)k^2 \leq 0$.

For the ordinary variants set old $J_3 = 2j$, old scale v_{old} as above, and restore the two centered source coordinates by

$$\tau_{\mathrm{old}} = 2\tau - (2u + k), \qquad y_{\mathrm{gap}} = y - V.$$

The first and auxiliary parent residuals become exactly

$$\tau_{\mathrm{old}}^2 + 2\tau_{\mathrm{old}}(2u + k) + k^2 - 4 = 4(\tau^2 - A_\tau),$$
$$V^2 + (1 - Q)y_{\mathrm{gap}}(2V + y_{\mathrm{gap}}) - 1 = QV^2 - (Q - 1)y^2 - 1.$$

The other three rows are the main norm, S and G. Every remaining parent abbreviation agrees literally. This restores the five-row signed CRT source directly, without inventing a new rank lemma. For normalized thirteen variables instead set

$$r_{\mathrm{old}} = \frac{\Delta(2r + 1) - 1}{2}, \qquad T_{\mathrm{old}} = fT.$$

The first quotient is integral because Δ is odd. Then $I_{\mathrm{old}} = \Delta I$, the parent's V agrees exactly, and its strong row follows from $f^2 = A_f$.

Conversely, the inherited canonical refutation witnesses have odd $J \equiv 1 \pmod 3$, so $J = 6j + 1$. Put $v = q^2(2v_{\mathrm{old}} + 1)$ and retain their guard witnesses. For example the ordinary fourteen-variable construction chooses

$$n = \gamma + 3\tau + y + 9p_1 - V - e/M.$$

The centered fifteen-variable construction chooses

$$n = 6\tau + 2y + 2\gamma - 3t_0 - 2V - e/M.$$

The other formulas omit the coordinates not fused by their norms. All values are integers and make one conjugate factor vanish. The normalized thirteen-variable case uses the same canonical odd strong normalization and fresh auxiliary reconstruction proved in the incoming prime-separation package. Its ordinary calibration boundary is unchanged.

1.6 Universal degree bounds and actual coefficient certificates

Fix the six ports. The exact propagated prefix bounds are

register	H	k	u	A_τ	A_g	Q	V	M	U	S, D
degree	4	5	11	22	28	6	7	2	6	10

in the ordinary case, with $\deg G = 28$. The normalized case instead has $\deg A_f = 14$, $\deg Q = 22$ and $\deg V = 19$. Direct degree propagation through the exact DAGs proves the final upper bounds $378, 172, 98$ for ordinary fourteen, ordinary fifteen and centered sixteen variables. Two cancellation-sensitive cases require more.

Thirteen variables. Pair the two signs of the auxiliary root first. Write

$$Z = UV + s, \qquad s = Un + Mb_0 + H(v - q^2),$$

and, for each of the sixteen sign choices, put

$$B = s \pm 9Uf \pm 3U\tau \pm Md_0 \pm 27Up_1.$$

Assign formal root weights $7, 11, 5, 14$ to f, τ, d_0, p_1. Each squared radicand has degree at most twice its root weight, so reduction in this formal root algebra never increases weighted degree. Thus B has weight at most 20. The paired factor is exactly

$$\delta\{(UV + B)^2 - U^2 C_y/\delta\} = \delta B^2 + 2\delta UVB - U^2(V^2 - 1).$$

The identity uses $C_y = (\delta + 1)V^2 - 1$. Its three terms have bounds $62, 67, 50$. The sixteen-factor product therefore has degree at most $16 \cdot 67 = 1072$, and is exactly the cleared five-root norm. This argument is polynomial algebra, including all exceptional inputs.

Centered fifteen variables. Use the exact centered two-root identity with its scale M replaced by U, and its carrier argument replaced by

$$x = Un + 2Mb_0 + H(v - q^2) \pm 2Md_0.$$

With d_0 of weight five, x has weight at most seven. Define

$$\begin{aligned} j &= \delta\{x^2 + U(6t_0 + 4V)x + 9U^2\ell\} - 4U^2(V^2 - 1), \\ h &= 12\delta t_0 VU^2. \end{aligned}$$

Their weights are at most 30 and 36. Every term of

$$j^2 + 2hj + 576\delta U^4 A_\tau(1 - V^2) + 144\delta^2 U^4 \ell V^2$$

has weight at most 66. The exact identity uses only $t_0^2 = 4A_\tau + \ell$, not a Pell zero. Taking the two main-root conjugates gives norm degree at most 132; the strong and guard finalizers add $10 + 28$, yielding 170.

Independent modular-line certificates modulo 65521 and 65519, using the actual six numeral byte strings, attain every stated bound with a nonzero leading coefficient. They are rigorous lower certificates combined with the preceding universal upper proofs. All literal DAGs pass topology, variable closure, liveness and coefficient hash checks. An independent exact sparse-algebra audit verifies every prefix, carrier, radicand, norm and finalizer, using all 20 349 coefficients of the generic five-root norm; deliberate corruptions are rejected. Signed modular diagnostics additionally include $c = 0$, $v = 0$ and their intersection. These algebraic checks do not establish an unbounded compiler theorem: the named ordinary source calibration remains an explicit mathematical dependency.